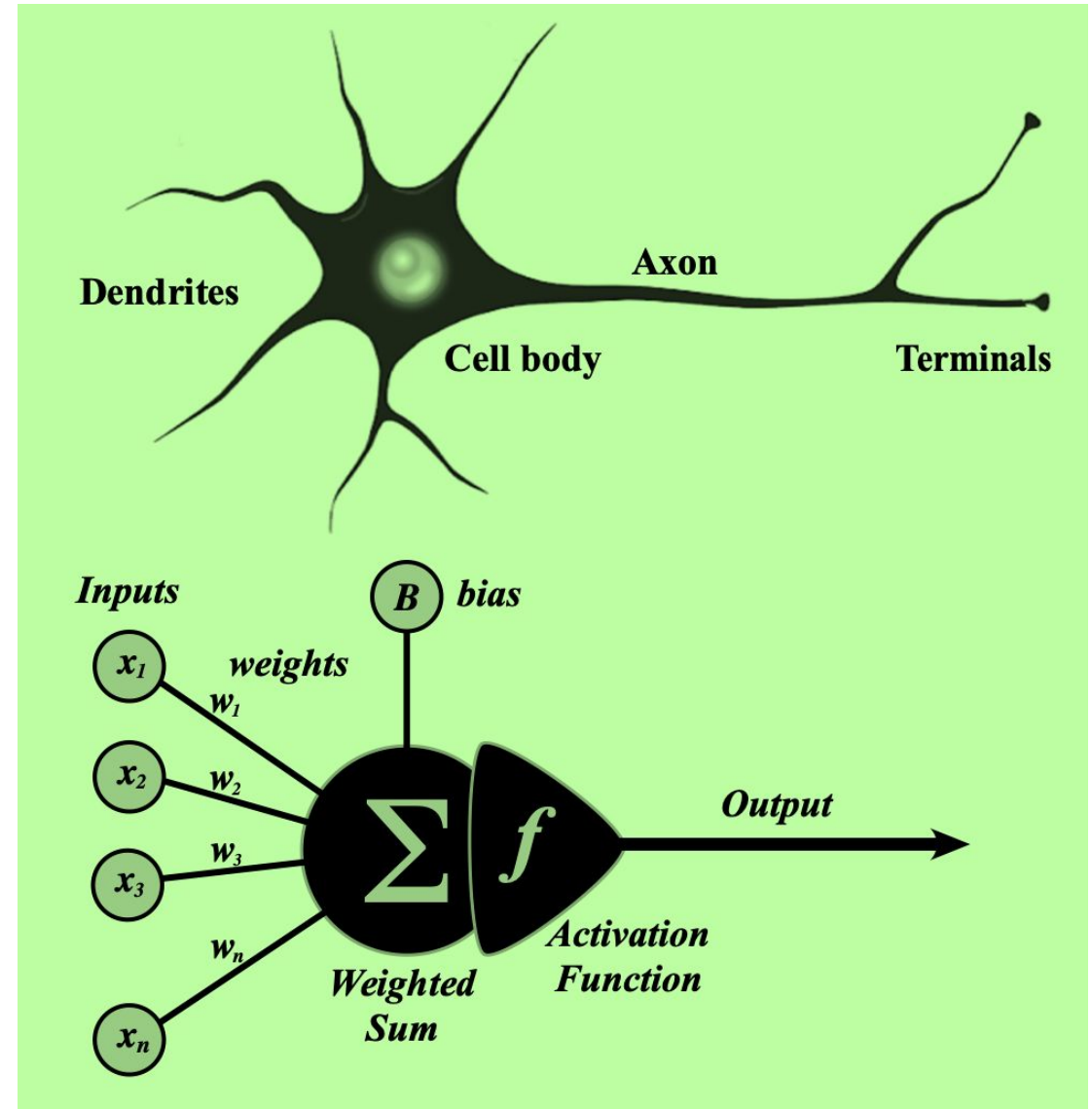


Artificial Neural Network

Dept. of CSE, University of Asia Pacific

Artificial Neural Network (ANN)

- Inspired by Biological Neurons –
Mimics the human brain's
interconnected neurons to process
complex data.



Single Neuron / Perceptron

Basic Structure:

- **Inputs:** $X=[x_1, x_2, \dots, x_n]$
- **Weights:** $W=[w_1, w_2, \dots, w_n]$
- **Bias:** b
- **Activation Function:** ϕ (e.g., Step, Sigmoid, ReLU)

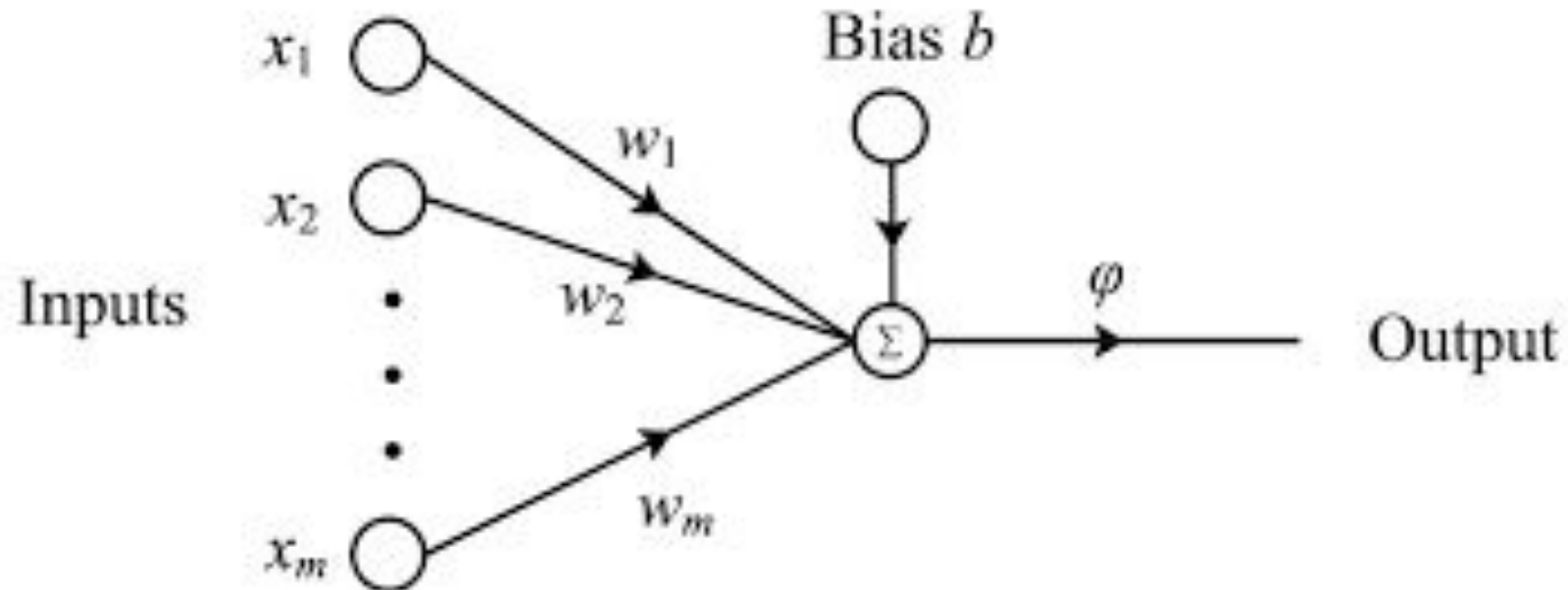
Single Neuron / Perceptron

- Pre-activation step:

$$z = \sum_{i=0}^n w_i x_i + b = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b$$

- Activation function:

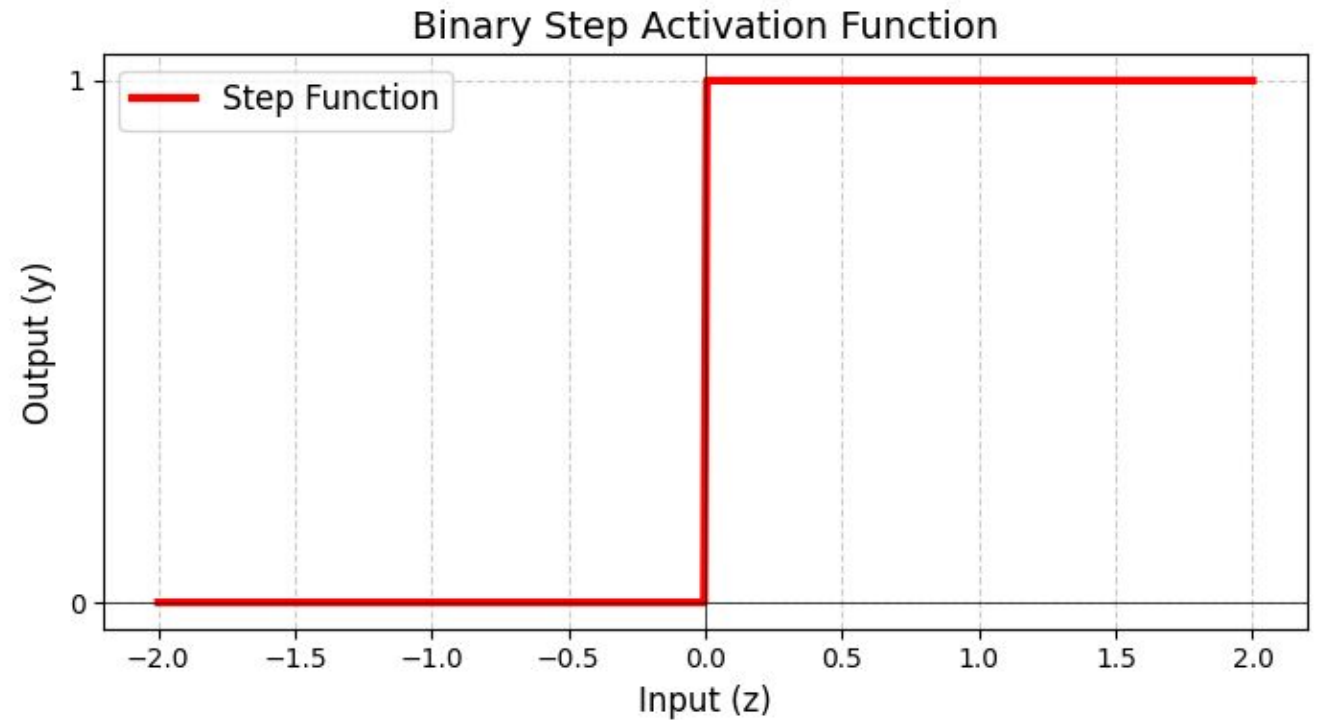
$$y = \phi(z)$$



Activation Functions

Binary step

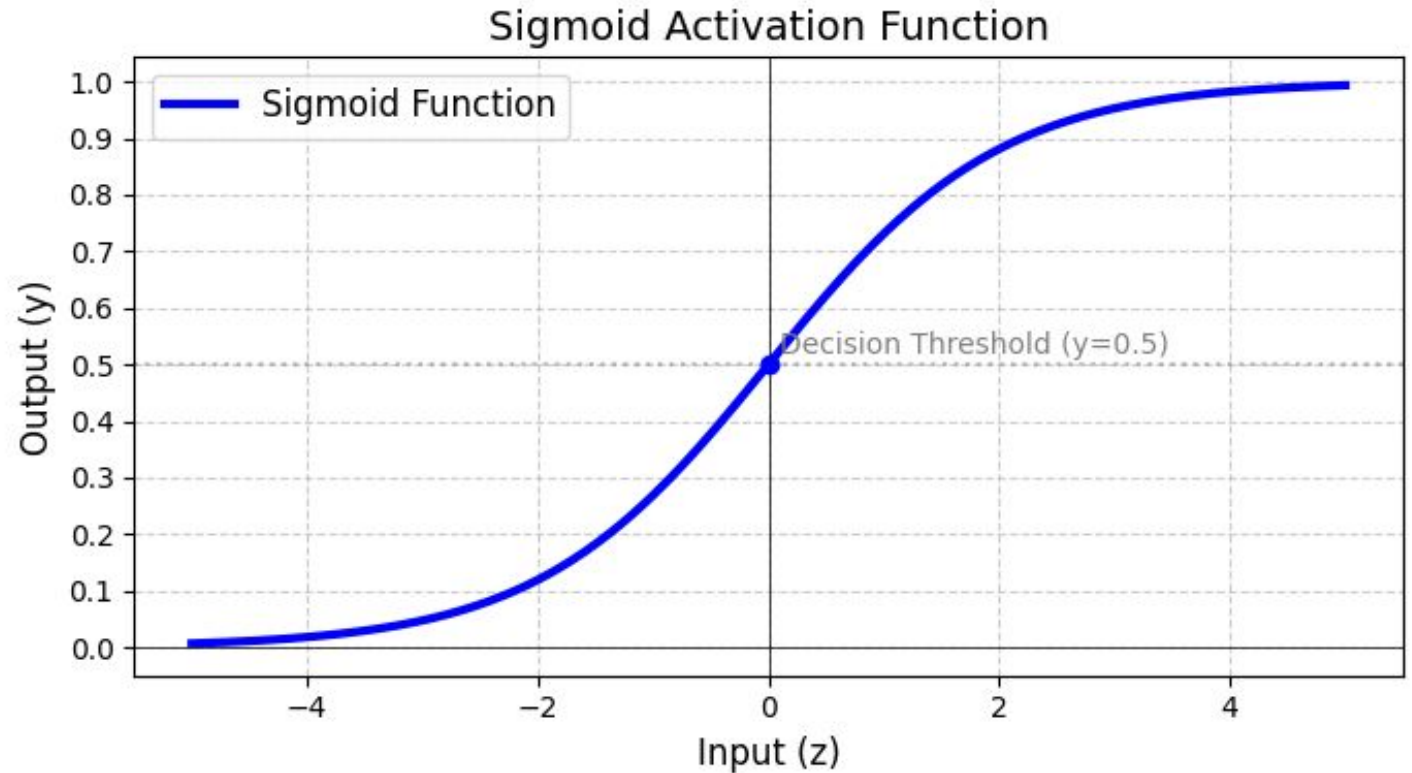
$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$$



Activation Functions (Cont.)

Sigmoid / Logistic

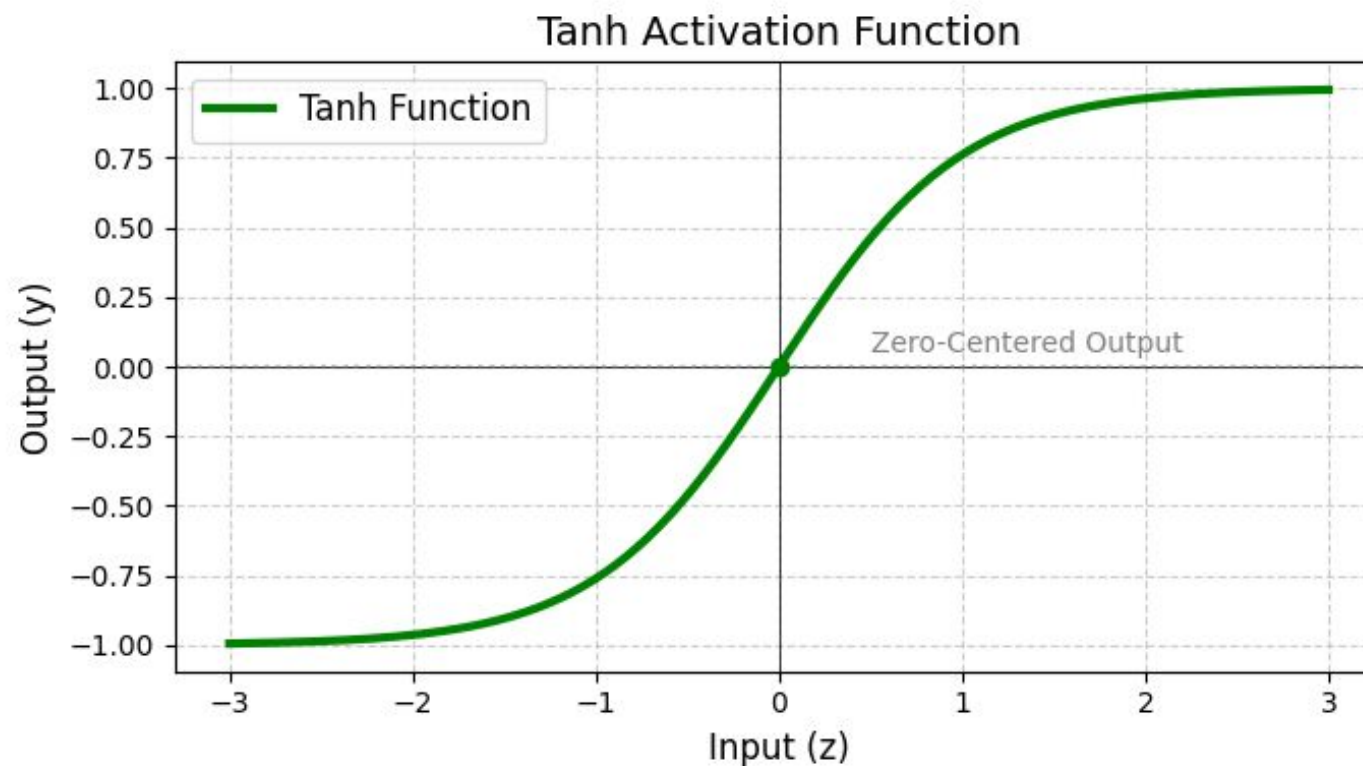
$$f(x) = \frac{1}{1 + e^{-x}}$$



Activation Functions (Cont.)

Tanh

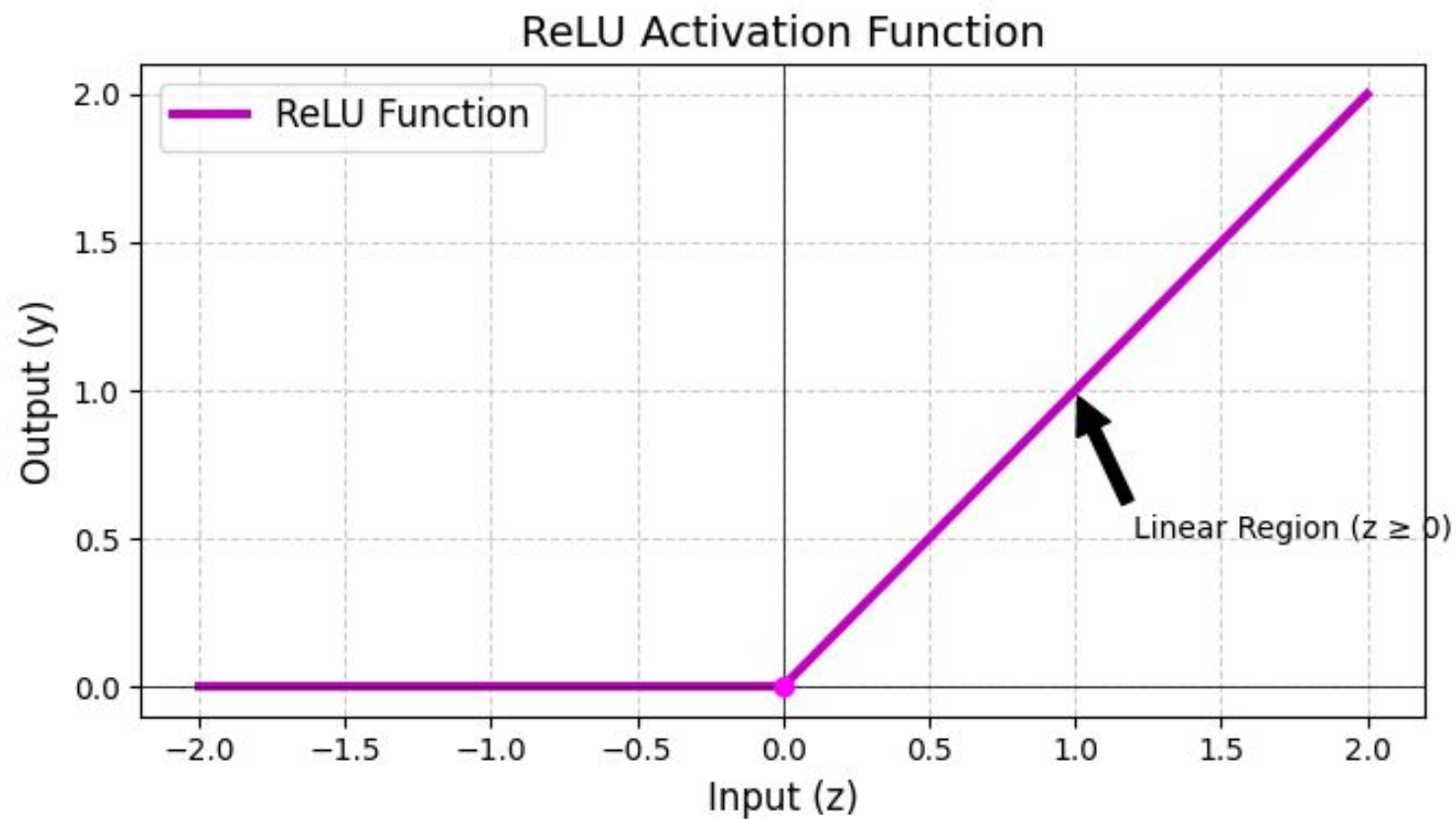
$$f(x) = \frac{(e^x - e^{-x})}{(e^x + e^{-x})}$$



Activation Functions (Cont.)

ReLU

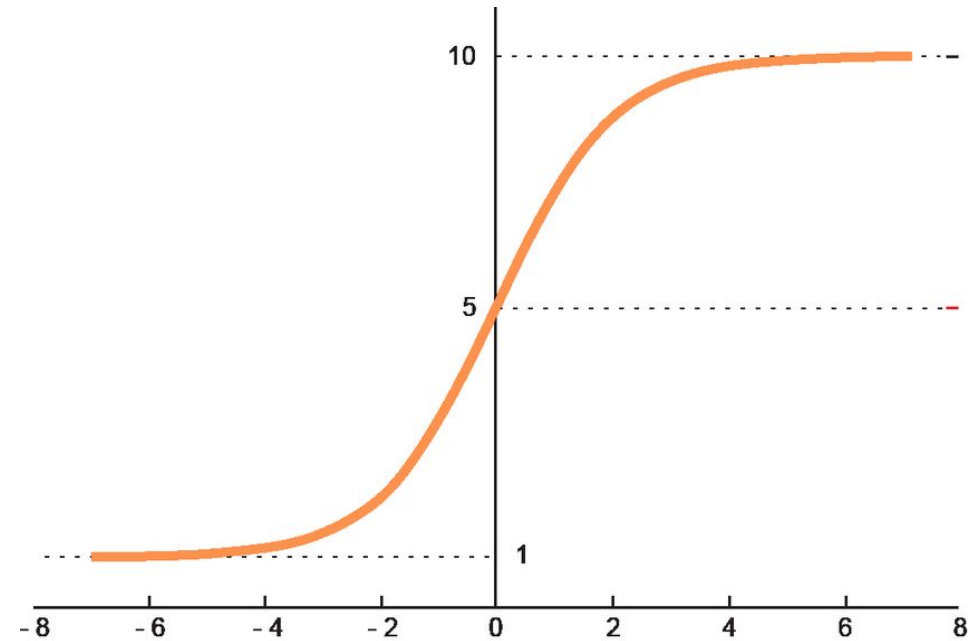
$$f(x) = \max(0, x)$$



Activation Functions (Cont.)

Softmax

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$



Perceptron Learning Rule

- Error, $E = Target - Output$
- Update weights & bias (if there is an error):

$$\Delta w_i = w_i + \alpha E x_i$$

$$b_{new} = b_{old} + \alpha E$$

where α is the learning rate.

Perceptron Learning Example: AND Gate

- Goal: to mimic AND logic, $y = x_1 \cdot x_2$
- Training data:

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

AND Gate: Initialization

- Initialize parameters (randomly)
 - Weights, $w_1 = 0.5$, $w_2 = 0.5$
 - Bias, $b = -0.3$
 - Learning rate, $\alpha = 0.1$

AND Gate: Training Steps

- Epoch 1:
 - Sample (0, 0)
 - $z = (0.5 \cdot 0) + (0.5 \cdot 0) - 0.3 = -0.3$
 - $y = \text{Step}(-0.3) = 0$
 - Error ($t - y$): $0 - 0 = 0$ $\square$ No update.
 - Sample (0, 1):
 - $z = 0.2, y = 1, \text{Error} = 0 - 1 = -1$ $\square$ Update
 - $W1 = 0.5$ (no change as $x1 = 0$)
 - $W2 = 0.5 + 0.1 \times (-1) \times 1 = 0.4$
 - $b = -0.3 + 0.1 \times (-1) \times 1 = -0.4$
 - Sample (1, 0)
 - $z = 0.1, y = 1, \text{Error} = -1$ $\square$ Update
 - $W1 = 0.4, W2 = 0.4, b = -0.5$
 - Sample (1, 1)
 - $z = 0.3, y = 1, \text{Error} = 0$ $\square$ No update

AND Gate: Training Steps

- Epoch 2:

- Weights now: $w_1 = 0.4$, $w_2 = 0.4$, $b = -0.5$
- $(0,0)$: $z = -0.5 \rightarrow y = 0$
- $(0,1)$: $z = -0.1 \rightarrow y = 0$
- $(1,0)$: $z = -0.1 \rightarrow y = 0$
- $(1,1)$: $z = 0.3 \rightarrow y = 1$

All predictions correct $\rightarrow$ Convergence achieved!

- Decision Boundary: $0.4x_1 + 0.4x_2 - 0.5 = 0$

Try with a different settings!

- Weights, $w_1 = 0.25$, $w_2 = 0.25$
- Bias, $b = -0.8$
- Learning rate, $\alpha = 0.5$

Questions?